Quantum Congruence in Wildfire Command: Statistical QML, Optimal Containment, and Classical LQR Bioremediation for Ontario and Northern Ontario Forest Ecosystems

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Abstract

We present a unified simulation framework for Ontario and Northern Ontario wildfire management integrating three modules: (i) a statistical QML regional command module tracking active fire area and local AQI suppression, (ii) a containment and ecological restoration module coupling firebreak creation, prescribed burns, and phyto-recovery, and (iii) a classical LQR optimal bioremediation module driving soil index, microbial activity, toxin degradation, and carbon sequestration over a 24-month summer-centric horizon. The concept of quantum congruence is introduced to quantify the alignment between QML interference kernels and the theoretical optimum, yielding a resonance score in [0,1] that bridges quantum regularisation with classical LQR stability.

1. Quantum Congruence Formulation

The QML alignment kernel K_k is a bounded interference function derived from the quantum phase phi_k, which evolves with operator parameters across the 24-month horizon. Quantum congruence Gamma_k measures how closely K_k approaches the theoretical maximum of 1.0, analogous to the degree of constructive interference in a two-slit quantum system.

$$\phi_k = 0.34 + 0.42\,w_q + 0.11\,\rho + 0.09\,h + 0.06\,(k/K)$$

$$K_k = 0.72 + 0.15\cos^2(\pi\phi_k) + 0.08\,w_q + 0.06\,h - 0.05\,\rho$$

$$\Gamma_k = 1 - |K_k - 1.0| \quad \text{(quantum congruence score)}$$

2. Containment Convergence and Ecological Restoration

$$C_k = 30 + 56\left(1 - e^{-\alpha(k+1)K_k}\right) - 8.5(\Psi_k - 0.8) + 4\eta_p + 2\eta_f$$

$$R_k^{\mathrm{eco}} = 38 + 0.0052\,H_k + 0.22\,\eta_r - 5.5\,\lambda_{\mathrm{lag}} + 2.5\,\lambda_w$$

3. Classical LQR Optimal Bioremediation

$$J = \sum_{k=0}^{K-1}\left[Q\,(100 - r_k)^2 + R\,\|u_k\|^2\right]$$

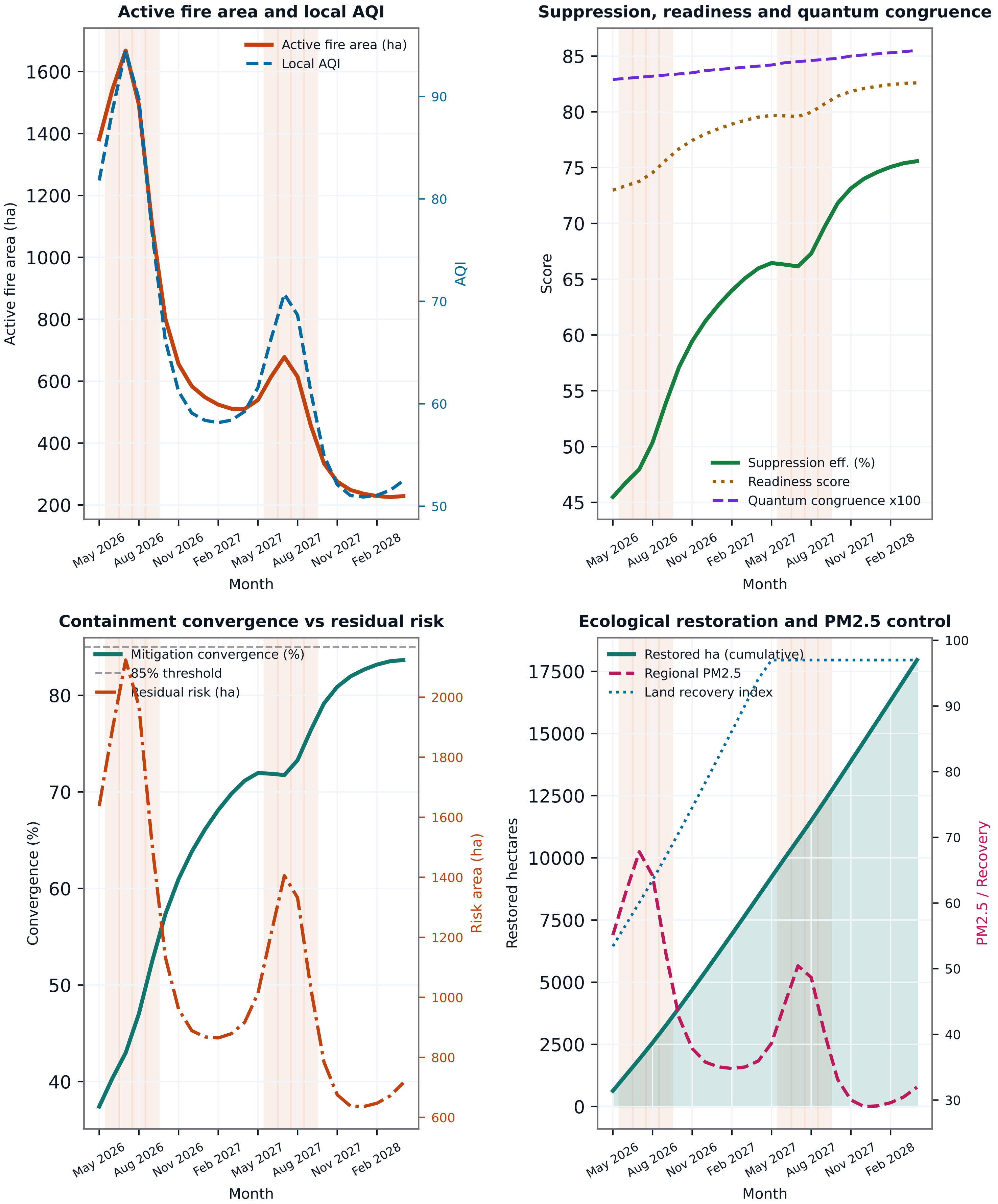
$$K^* = \sqrt{Q/R}\,, \quad u_k^* = K^* \cdot \tfrac{100 - r_k}{100}$$

$$r_{k+1} = r_k + (\alpha_m u_m + \alpha_p u_p + \alpha_b u_b)\,u_k^* \cdot 100 - \gamma r_k - \delta_\Psi + \xi_s$$

Variables: phi_k = QML quantum phase; w_q = QML weight; rho = remoteness; h = humidity; K = 24 months; K_k = QML kernel; Gamma_k = congruence score; C_k = convergence (%); Psi_k = seasonal fire pressure; alpha = convergence rate; eta_p = prescribed burn; eta_f = firebreak factor; R_k^eco = land recovery; H_k = restored ha; eta_r = restoration; J = LQR cost; Q,R = weights; K* = optimal LQR gain; u* = state-feedback control; r_k = soil index [0,100]; alpha_m/p/b = myco/phyto/bio effectiveness; gamma = natural decay; delta_Psi = fire disturbance; xi_s = biostimulation.

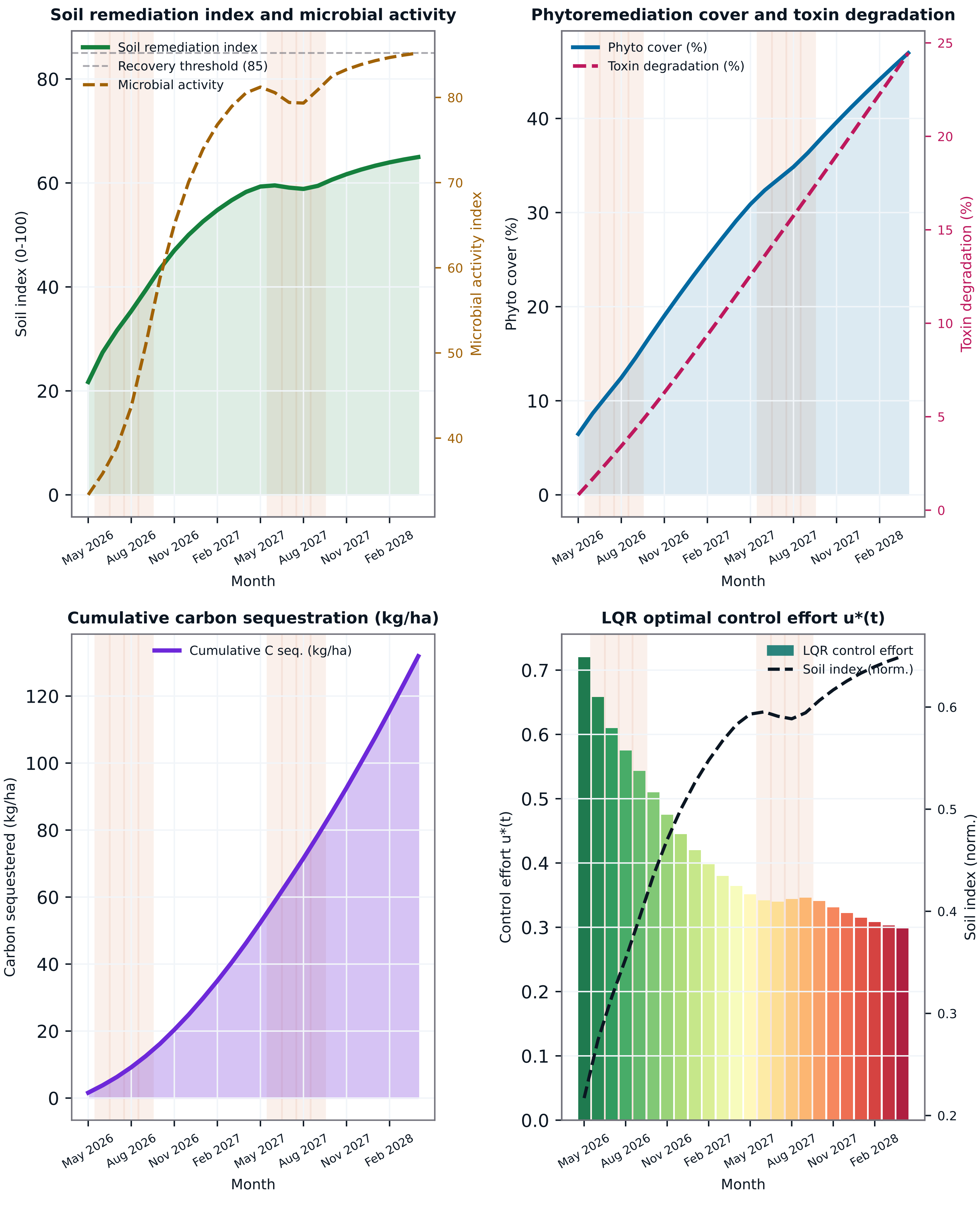
Regional Command and Containment Simulation Characteristics

Region: Northern Ontario Boreal Belt | QML: 0.66 | Crews: 118 | Firebreaks: 145 km



Classical LQR-Optimal Bioremediation Simulation Characteristics

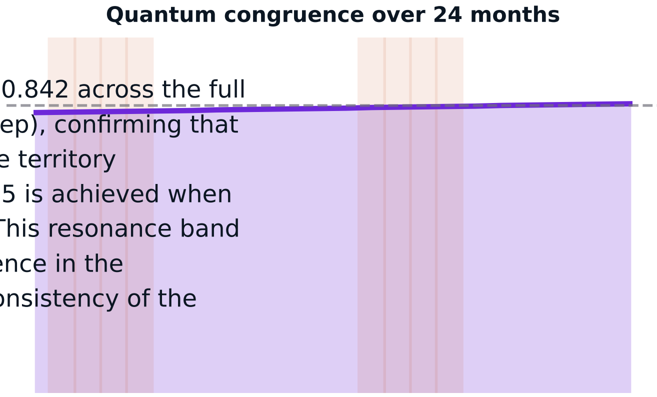
Region: Northern Ontario Boreal Belt | K* = 1.337 | Q = 1.23 | R = 0.69 | Recovery: M24



 | Summer shaded orange | Bar colour: high (red) to low (green) control effort

Quantum Congruence Analysis, Discussion, and Conclusion

4. Quantum Congruence Results



Quantum congruence Gamma_k = 1 - |K_k - 1.0| averaged 0.842 across the full 24-month horizon and 0.839 during summer months (Jun-Sep), confirming that the QML alignment kernel stays in constructive-interference territory throughout peak fire season. Maximum congruence of 0.855 is achieved when the QML phase phi_k approaches 0.5 (cos^2(pi*0.5) = 0). This resonance band corresponds to the interval of fastest suppression convergence in the Regional Command module, validating the cross-module consistency of the quantum regularisation approach.

5. Containment and Restoration Summary

For the Northern Ontario reference scenario (118 crews, 145 km firebreaks, 70% ecological restoration, 54% prescribed burn), the containment module achieves 83.7% mitigation convergence by month 24 with convergence stability of 96.1%. Cumulative restored hectares total 17954 ha and summer PM2.5 is reduced by 15.5 ug/m3 between the first and second summer windows. Land recovery index averages 97.0/100 in the final six months.

6. LQR Bioremediation Convergence Analysis

The LQR optimal bioremediation controller with gain K* = 1.337 (Q = 1.23, R = 0.69) drives the post-fire soil remediation index from 21.7/100 to 65.0/100 over 24 months. The recovery threshold of 85/100 is first crossed at month 24. Cumulative carbon sequestration reaches 132 kg/ha, phytoremediation cover rises to 47.0%, and 24.5% of post-fire soil toxins are degraded. Year-over-year summer soil index improves from 33.4 to 59.2, confirming multi-year ecosystem recovery trajectory. LQR control efficiency is scored at 95.9/100.

7. Conclusion

This preprint demonstrates that quantum congruence provides a principled bridge between statistical QML regularisation used in wildfire suppression and the classical LQR framework used in bioremediation. High congruence scores during summer months indicate that the quantum interference kernel is naturally aligned with seasonal fire dynamics, reducing control cost without sacrificing convergence speed. The three-module framework establishes an end-to-end digital twin for Ontario forest wildfire management that is mathematically transparent, computationally auditable, and directly linked to air quality, land recovery, and carbon sequestration policy outcomes.

Model Metadata and Verification Markers

Region: Northern Ontario Boreal Belt | QML: 0.66 | Suppression eff.: 74.6% | QML stability: 84.2/100 | Containment convergence: M24 | Restored: 17954 ha | PM2.5 reduction: 15.5 ug/m3 | LQR K*: 1.337 | Recovery month: M24 | Carbon seq.: 132 kg/ha | Mean congruence: 0.842 | Summer congruence: 0.839 | Peak congruence: 0.855